Apple SDS Digital Goods
Technical Fraud Analysis

Dashboard Overview: Apple SDS Digital Fraud Analysis

Welcome to the Apple SDS Digital Fraud Analysis Dashboard. This dashboard provides a comprehensive, real-time view into Account Takeover (ATO) fraud patterns and the performance of our automated detection and decisioning system within Apple's Services segment.

Leveraging enriched data from Google BigQuery, this tool is designed to empower fraud analysts with actionable insights, enabling effective monitoring of key performance indicators (KPIs), identification of emerging fraud trends, and assessment of the system's 'friction-right' balance—minimizing false positives while maximizing fraud capture.

Key Features:

Executive Summary: High-level metrics on overall fraud rates, system block/review rates, and estimated revenue protected.

User & Device Insights: Detailed views into suspicious user behaviors and device sharing patterns.

Geographical Fraud Analysis: Visual identification of fraud hotspots based on transaction IP locations and cross-country activity.

This dashboard serves as a critical operational tool, directly supporting the mission to protect digital goods transactions and maintain a seamless, secure user experience.

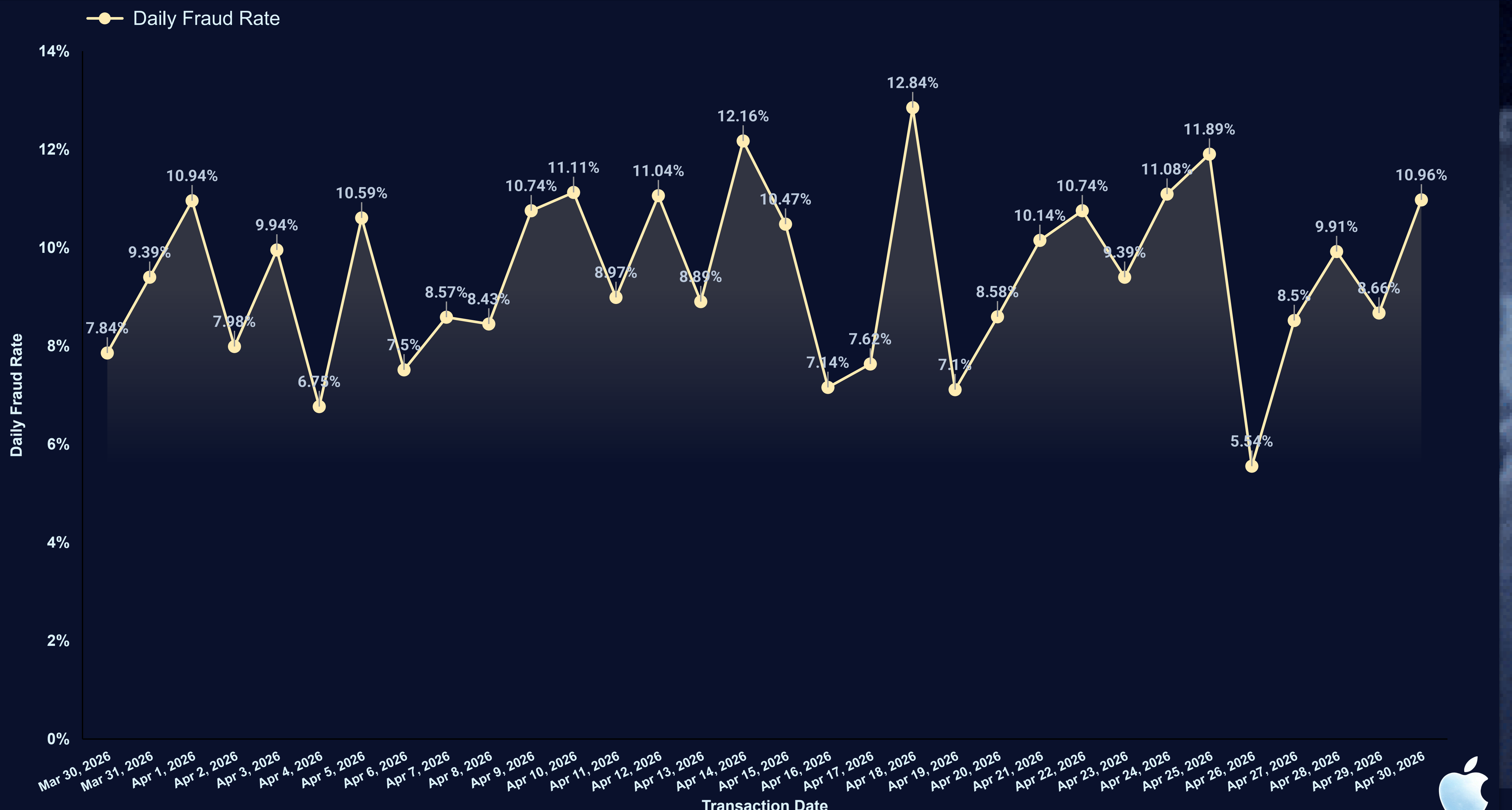
Fraud Analysis
Portfolio Project
Daniel Rodriguez III
30 April 2026

OVERVIEW & KEY METRICS

Overall Fraud Rate	System Block Rate	System Review Rate	False Positive Rate	Estimated Revenue Protected
9.44%	9.57%	4.57%	0.14%	$104,102.65

This time series chart visualizes the daily fluctuations in the fraud rate, helping to identify trends, seasonal patterns, or sudden spikes in fraudulent activity. An upward trend might signal a need for immediate investigation and system adjustments.

Daily Fraud Rate Over Time



USER & DEVICE INSIGHTS

User ID | Total Spend | Device ID | Distinct IPs

This table identifies the users associated with the highest number of fraudulent transactions and their total spending. Highlighting these users can guide investigations into potential fraud rings or compromised accounts.

Top Users by Fraudulent Transactions & Total Spend

	user_id	Fraudulent Transactions	Total Spend
1.	user_738	10	$1,515.67
2.	user_170	9	$1,364.62
3.	user_8	7	$1,044.65
4.	user_710	7	$983.82
5.	user_491	7	$833.15
6.	user_165	7	$512.48
7.	user_598	6	$1,320.53
8.	user_902	6	$1,043.34
9.	user_481	6	$905.95
10.	user_505	6	$823.18

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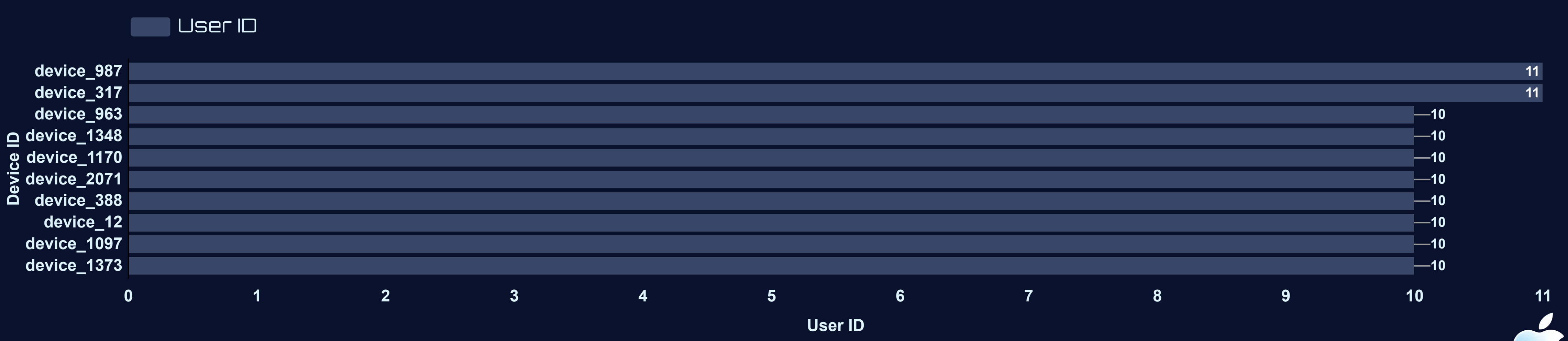
This bar chart displays device IDs used by the highest number of distinct users. A high count of unique users sharing a single device ID could indicate account sharing, mule accounts, or sophisticated fraud operations using emulators.

Excessive Device Sharing

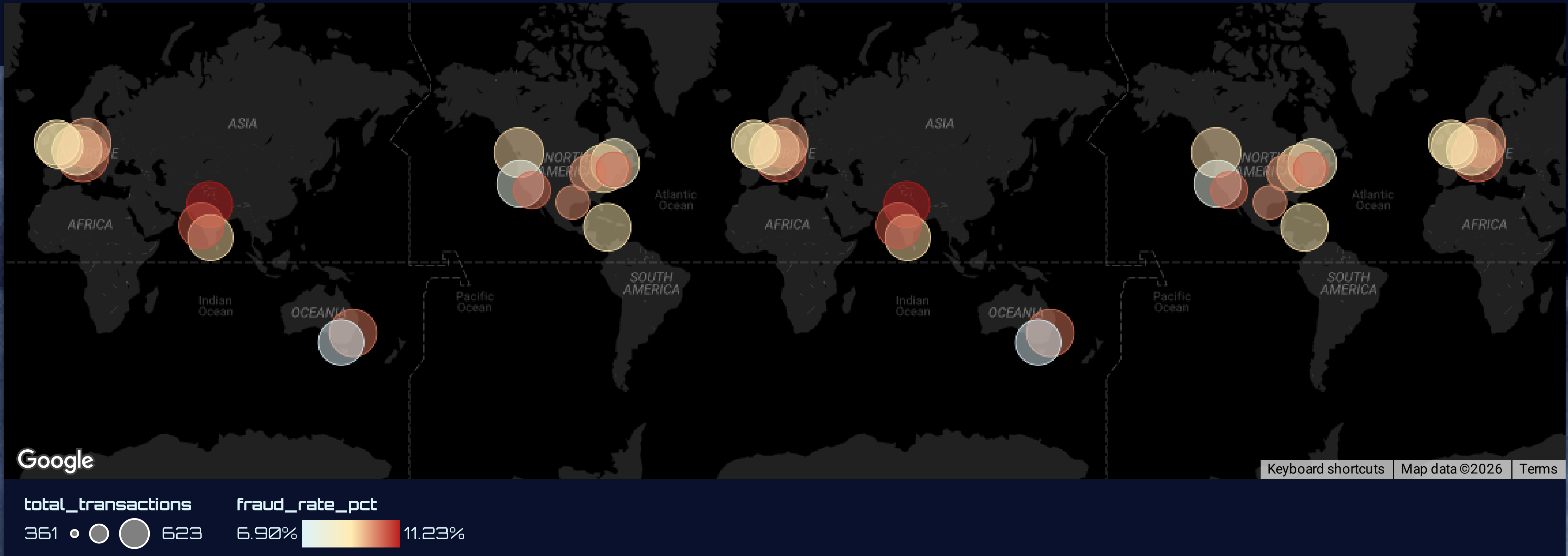


GEOGRAPHICAL FRAUD ANALYSIS

Transaction IP Country | IP City | Transaction ID | User ID | Is Compromised | Total Spend

Fraud by IP Country/City

This Bubble Map visualizes the fraud rate by IP country and city, highlighting geographical hotspots where fraudulent activities are more prevalent. This insight is crucial for implementing region-specific fraud rules or focusing investigative efforts. The drill-down functionality allows for a more granular view from country to city level.



Cross-Country Transactions Analysis

This table analyzes established cross-country transaction corridors, mapping the flow from a user's registered country to the transaction's origin IP. By filtering out low-volume outliers to ensure statistical significance, the analysis isolates high-risk geographical 'hotspots' where disparate locations serve as a validated indicator of fraud. This data is critical for refining location-based security rules and focusing investigative resources on the most impactful fraudulent paths.

	user_registered_country	Country Corridor	transaction_ip_country	Transaction Count	Fraud Rate
1.	IN	IN → AU	AU	18	11.11%
2.	CA	CA → IN	IN	14	0.00%
3.	US	US → DE	DE	14	7.14%
4.	DE	DE → AU	AU	13	7.69%
5.	AU	AU → UK	UK	13	15.38%
6.	AU	AU → CA	CA	13	7.69%

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